

SUBJECT

Idealized trileaflet aortic valve, systolic configuration

OBJECTIVE	Optimize the leaflet opening angle of the aortic valve to minimize pressure loss over the cardiac cycle. Decompose the cycle into representative phase points, rule on the admissible method, and report the cycle-weighted loss with its uncertainty.
SOLVER & MODEL	Reduced-order orifice model, cycle-decomposition screen
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RESULT

CONCEPTUAL MODEL

Cycle-weighted pressure loss

1345 Pa ± 441 Pa

95% confidence interval

Basis: reduced-order orifice screen on a screening geometry; the flow is inertially unsteady

Winning leaflet opening 80 deg orifice 403 mm2

UNCERTAINTY - ASME V&V 20 THREE-CHANNEL

CHANNEL	VALUE	STATE
input	421.1424	quantified
2-sigma Monte-Carlo envelope propagated from the spread in phase flow-rate and the orifice discharge coefficient		
numerical	81.0	quantified
phase-quadrature ladder k = 3/5/9		
model	104.8	quantified
correlation-family spread; unmodeled: phase-interaction neglected, leaflet motion not modeled, Newtonian blood		

PROVENANCE - EVIDENCE-BUNDLE SEAL (SHA-256)

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Mission valve-study - Certificate C-2026-8945